

Apply for a Science Corps Fellowship - Entry #3236

Name

William Chen

Emailwchen23@nd.edu**Gender**

Male

City, state, and country of residence

West Covina, CA 91790

Nationality

United States

When did or will you finish your PhD?

6/15/2024

University Education

University of California, Los Angeles (UCLA) - Bachelor's of Science Degree in Civil and Environmental Engineering (completed)

Stanford University - Master's of Science Degree in Civil and Environmental Engineering (completed)

University of Notre Dame - Doctorate Degree (PhD) in Civil and Environmental Engineering and Earth Sciences (ongoing)

What are your reasons for applying to be a Science Corps fellow?

I've always stood apart from the crowd. Unlike many, I don't aspire to amass wealth, seek fame, or work for a tech giant producing endless products to fuel a lavish lifestyle. While these goals are grand, I find them lacking in nobility, offering no solutions to the world's most pressing issues. How can I indulge in opulence while countless individuals, including vulnerable children and the elderly, must bear the negative impacts of hunger, poverty, diseases, natural disasters, and more? My purpose in life, I believe, lies in serving society and addressing the needs of the most disadvantaged. This

stems from my own journey as a first-generation student who faced formidable socioeconomic obstacles. Thus, I'm driven to pursue a greater mission, one that extends beyond personal gratification and aids those residing in underdeveloped regions.

Education, particularly in the scientific realm, stands as a crucial pathway to surmount daily challenges. I, too, owe my current position to countless hours spent in a public library, learning mathematics, chemistry, and physics on my own. This self-education equipped me with the skills, knowledge, and confidence to navigate rigorous university courses and beyond. This journey was arduous, as I lacked guidance, resources, or personal tutors to help me due to my family's financial constraints. Now, envisioning the struggles of those in even more underprivileged circumstances, I am passionate about leveraging education as a catalyst for change. This teaching abroad opportunity offered by Science Corps perfectly aligns with my aspirations, offering a chance for me to make a meaningful difference in the lives of those in need.

In addition to your studies, describe any relevant experiences that have prepared you for a Science Corps fellowship.

Professionally, I possess teaching and tutoring experience across diverse student backgrounds. For instance, I mentored my cousin Lawrence in mathematics and physics during his high school years, equipping him with the confidence to conquer more advanced courses at a prestigious institution like Johns Hopkins University. Additionally, I contributed to the education of naval officers in mathematics and science during my tenure in the US Navy. Finally, I served as a teaching assistant at both Stanford University and the University of Notre Dame. In this capacity, I conducted office hours and facilitated discussion sessions, enhancing students' problem-solving skills and their holistic understanding of subjects in the course beyond how they were taught in class. This helped students look at problems from a different angle that will challenge them to become better future problem-solvers.

Moreover, my international experiences, including studying in Ecuador, have broadened my perspective. Upcoming research trips to Trinidad and Tobago and Kenya will allow me to further immerse myself in underprivileged communities, comprehend their cultural dynamics, and witness the challenges and opportunities they face. These encounters reaffirm my commitment to helping these communities overcome barriers, such as gender inequality and resource limitations, to access education and self-improvement.

In summary, my diverse teaching background and international exposure position me well to excel in the Science Corps teaching abroad opportunity.

Where do you imagine yourself professionally in the future? Check all that apply.

Academic scientist in academia

Doing science in industry

In science outreach

Expand on your selection above and describe where you imagine yourself professionally in the near (two year) and long-term (ten year) future.

I aspire to become a professor, merging my research interests with educating future students and like-minded researchers over the next decade and beyond. The academic setting allows me to delve into in-depth analysis and innovative problem-solving, while teaching brings me profound fulfillment, especially when I discover unique approaches to engineering and mathematical challenges.

My primary ambition revolves around utilizing research findings to address issues confronting underserved communities in environmental and public health domains. My current research combines model-based techniques with hands-on experimentation to monitor diseases in wastewater. This approach aims to enhance comprehension and practical applications, potentially transforming the way infectious diseases are tracked and mitigated in developing regions.

Recognizing the collaborative nature of successful research, my aspirations extend beyond personal achievements to mentor upcoming researchers, equipping them with essential tools for success. I am driven to inspire others to engage in research that advances the well-being of underserved communities. Hence, this teaching opportunity abroad serves as a good starting point for me to refine my teaching skills while remaining committed to my ultimate goal of contributing to resource-limited areas through science education. It will also offer valuable firsthand experience in underprivileged countries, preparing me for future research initiatives in similar global contexts. I envision dedicating the next six months to this endeavor as part of my preparation for a post-doctorate position at a university for the following two years, all contributing to my long-term goal of becoming a professor devoted to researching ways to uplift underprivileged communities.

CV/Resume

 [William-Chen-CV-9 13 2023.pdf](#)

How did you find out about Science Corps?

University email list

Any other comments/questions?

Empty

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